

MAT9004 Assignment - 2

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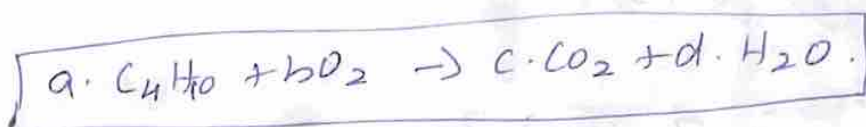
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1) Given unbalanced equation is as follows:



Given coefficient to balance the above equations are
a, b, c & d.

$\therefore$ The balanced equation is as follows.



a) System of equations:

Carbon:

There are 4 carbon for a coefficient &
1 carbon for c-coefficient.

$$\therefore 4a = 1c$$

$$4a - c = 0 \quad - (1)$$

Hydrogen:

There are 10 hydrogen for a coefficient &
2 hydrogen in d coefficient.

$$\therefore 10a = 2d$$

$$10a - 2d = 0 \quad - (2)$$



Oxygen:

There are 2 oxygen for b coefficient, 2 oxygen for c coefficient + 1 oxygen for d coefficient.

$$\therefore 2b = 2c + d.$$

$$\boxed{2b - 2c - d = 0} \quad - (3)$$



b) Matrix form:

To write (1), (2) & (3) in matrix form,

$$\begin{bmatrix} 4 & 0 & -1 & 0 \\ 10 & 0 & 0 & -2 \\ 0 & 2 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



c) Gaussian Elimination:

$$\left[\begin{array}{cccc|c} 4 & 0 & -1 & 0 & 0 \\ 10 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right]$$

$$R_1 = \frac{R_1}{4}$$

$$\Rightarrow \left[\begin{array}{cccc|c} \frac{4}{4} & \frac{0}{4} & \frac{-1}{4} & \frac{0}{4} & \frac{0}{4} \\ 10 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 0 & -0.25 & 0 & 0 \\ 10 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right]$$

$$R_2 = R_2 - 10R_1$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -0.25 & 0 & 0 \\ 0 & 0 & 2.5 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array} \right]$$

$$R_2 = R_3$$

$$R_3 = R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -0.25 & 0 & 0 \\ 0 & 2 & -2 & -1 & 0 \\ 0 & 0 & 2.5 & -2 & 0 \end{array} \right]$$

Now we perform substitution + keep all the values of a, b, c in terms of d .

$$2.5c - 2d = 0$$

$$\Rightarrow 2.5c = 2d$$

$$c = \frac{2d}{2.5}$$

$$\boxed{c = 0.8d}$$

$$2b - 2c - d = 0$$

$$2b - 2(0.8d) - d = 0$$

$$2b - 1.6d - d = 0$$

$$2b - 2.6d = 0$$

$$2b = 2.6d$$

$$b = \frac{2.6d}{2}$$

$$\boxed{b = 1.3d}$$

$$a - 0.25c = 0$$

$$a - 0.25(0.8d) = 0$$

$$a - 0.2d = 0$$

$$\boxed{a = 0.2d}$$

To clear all the decimal we can have $d = 5$.

$$\Rightarrow a = 0.2(5) = 1$$

$$b = 1.3(5) = 6.5$$

$$c = 0.8(5) = 4$$

$$d = 5$$

6.5 is not whole. $\therefore$
Multiply all the values by 2

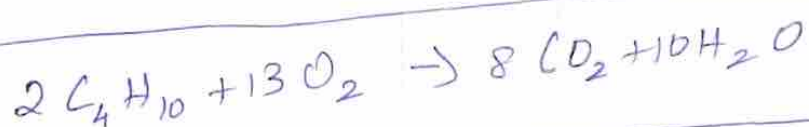
$$\boxed{a = 2 \mid b = 13 \mid c = 8 \mid d = 10}$$

∴ Solution in vector form is

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2 \\ 13 \\ 8 \\ 10 \end{bmatrix}$$

4.1

d) Balanced equation:



2) Given parabola

$$y = -x^2 + 4x - 3$$

a) Find tangents at $x = 0$ & $x = 3$.

at $x = 0$,

$$y = 0^2 + 4(0) - 3$$

$$y = -3$$

at $x = 3$,

$$y = -(3)^2 + 4(3) - 3$$

$$= -9 + 12 - 3$$

$$= 0$$

∴ points are $(0, -3)$ $(3, 0)$

To find slope, we need to find y'

$$\boxed{y' = -2x + 4}$$

$$\begin{array}{l|l} \text{slope at } x=0 & \text{slope at } x=3 \\ \boxed{y' = 4} & \boxed{y' = -2} \end{array}$$

$\therefore$ The tangent line has
point $(0, -3)$ | point $(3, 0)$
slope $(\frac{y}{x}) = 4$ | slope $= -2$

Using the slope point form,

$$y - y_1 = m(x - x_1)$$

$$y - (-3) = 4(x - 0)$$

$$y + 3 = 4x$$

$$~~y = 4x - 3~~$$

$$\boxed{y = 4x - 3}$$



$$y - 0 = -2(x - 3)$$

$$\boxed{y = -2x + 6}$$

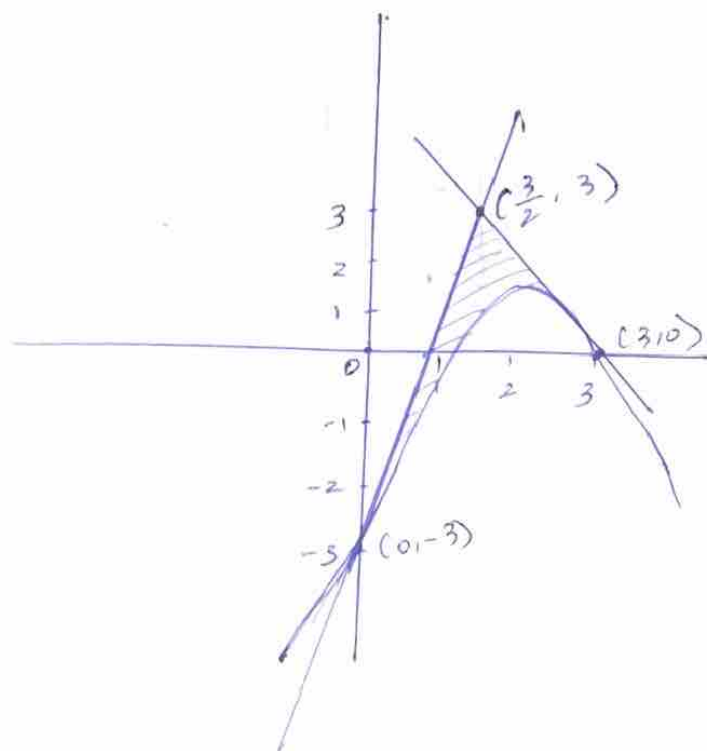
$$\boxed{y = -2x + 6}$$



$\therefore$ Equation of tangents at $x=0$ & $x=3$ are

$$\boxed{y = 4x - 3} \text{ \& \ } \boxed{y = -2x + 6} \text{ respectively.}$$

b) Sketch the parabola.



Find point of intersection of the two tangents.

Tangent:

$$y = 4x - 3$$

Parabola:

$$y = -x^2 + 4x - 3$$

$$4x - 3 = -x^2 + 4x - 3$$

$$x^2 = 4x - 3 - 4x + 3$$

$$x^2 = 0$$

$$\boxed{x = 0}$$

$$y = 4(0) - 3$$

$$\boxed{y = -3}$$

$\therefore$ Point of

Tangent to parabola

is $(0, -3)$

Tangent

$$y = -2x + 6$$

Parabola

$$y = -x^2 + 4x - 3$$

$$-2x + 6 = -x^2 + 4x - 3$$

$$x^2 = 4x - 3 + 2x - 6$$

$$x^2 = 6x - 9$$

$$x^2 - 6x + 9 = 0$$

$$x^2 - 3x - 3x + 9 = 0$$

$$x(x - 3) - 3(x - 3) = 0$$

$$(x - 3)(x - 3) = 0$$

$$\boxed{x = 3}$$

$$y = -6 + 6 = 0$$

$$\begin{array}{r} x \\ -3 \overline{) -3} \end{array}$$

$\therefore$ Point of

Tangent on parabola

is $(3, 0)$

Point of intersection of the two tangents

$$\Rightarrow 4x - 3 = -2x + 6$$

$$4x + 2x = 6 + 3$$

$$6x = 9$$

$$x = \frac{9}{6}$$

$$\boxed{x = \frac{3}{2}}$$

$$y = 4\left(\frac{3}{2}\right) - 3$$

$$\boxed{y = 3}$$

$\therefore$ Point of intersection of the two tangents is

$$\left(\frac{3}{2}, 3\right)$$



3)

a) Given that the rate at which water is released is at $4e^{-0.03t}$ megalitres per day.

Using calculator, as I found that

$$e^{-0.03} \approx 0.97044 \text{ which is } > 0.$$

$\therefore 4e^{-0.03t}$ will always be > 0 for all real t . 

The way we can understand is that the reservoir releases water continuously.

The rate of release of water reduces gradually as time goes by due to the demand. That is ~~at~~ initially the water demand will be high so the rate starts at a higher value. As the demand of the water is met, the rate of release gradually decreases.

This ensures the water is preserved for all seasons. 

b) Given:

$$\frac{dV}{dt} = 4e^{-0.03t}$$

$$V = \int 4e^{-0.03t} dt = 4 \cdot \frac{e^{-0.03t}}{-0.03} + C$$

$$= -\frac{4}{0.03} e^{-0.03t} + C$$

To find C , we let $t=0$ + $V(0)$

$$0 = -\frac{4}{0.03} e^0 + C$$

$$\boxed{C = \frac{4}{0.03}}$$

$$\therefore V(t) = -\frac{4}{0.03} e^{-0.03t} + \frac{4}{0.03}$$

$$\boxed{V(t) = \frac{4}{0.03} (1 - e^{-0.03t})}$$



c) When $t \rightarrow \infty$

$$e^{-0.03t} \rightarrow 0$$

$$\therefore V(\infty) = \frac{4}{0.03} (1 - e^0)$$

$$= \frac{4}{0.03} (1)$$

$$= \frac{400}{3} = 133.33 \text{ megalitres.}$$

$\therefore$ The total volume of the reservoir is 133.33 megalitres

When $t \rightarrow 60$

$$V(60) = \frac{4}{0.03} (1 - e^{-0.03(60)})$$

$$= \frac{4}{0.03} (1 - 0.1652)$$

$$= \frac{4}{0.03} (0.8348)$$

$$= 111.30 \text{ megalitres.}$$

$\therefore$ By the end of 60 days, 111.30 megalitres of water would have been released

~~111.30 meg~~

111.30 megalitres would equate to $\frac{111.30}{133.33} \times 100$

$\Rightarrow 83.47$

$\therefore$ 111.30 megalitres would equate to 83.47% of the total water released during the first 60 days



4)

a) Let x_n be the number of machines infected by malware x at time n .

Let y_n be the number of machine infected by malware y at time n .

Given:

with malware x ,

5 new machines with x

2 new machine with y .

itself stays infected.

$\therefore x$ contributes 5 (new) + 1 (itself) = 6 with x + 2 machines with y .

~~4 machines~~

With malware y ,

3 machines with x

4 machines with y

itself stays infected.

$\therefore y$ contributes 3 machines with x + 4 (new) + 1 (itself) = 5 machines with y .

$$x_{n+1} = 6x_n + 3y_n$$

$$y_{n+1} = 2x_n + 5y_n$$

In matrix form,

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}$$

where $A = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix}$



b) Given

$$M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

~~Find~~

$$M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Find M^{-1}

$$M^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where $a=4$ $b=1$ $c=2$ $d=3$

$$M^{-1} = \frac{1}{4(3)-1(2)} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{12-2} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$$

From a, we can write a equation such as

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = M \begin{bmatrix} x_n \\ y_n \end{bmatrix}$$

$$\Rightarrow x_{n+1} = M [x_n]$$

From question, $x_3 = 125$ & $y_3 = 125$.

$$\therefore \begin{bmatrix} 125 \\ 125 \end{bmatrix} = M \begin{bmatrix} x_n \\ y_n \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 125 \\ 125 \end{bmatrix} = M \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \left(\begin{array}{l} x_{n+1} = 3, \\ y_{n+1} = 2 \end{array} \right)$$

we need to find $\begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$,

$$\therefore \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = M^{-1} \begin{bmatrix} x_3 \\ y_3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 125 & 3 \\ 125 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 125 & 3 \\ 125 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

~~$= \frac{1}{10} (125(3) - 2(125))$~~

$$\Rightarrow \begin{bmatrix} \frac{3}{10} & -\frac{1}{10} \\ -\frac{2}{10} & \frac{4}{10} \end{bmatrix} \begin{bmatrix} 125 \\ 125 \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{10} \begin{pmatrix} 125 \\ 125 \end{pmatrix} - \frac{1}{10} \begin{pmatrix} 125 \\ 125 \end{pmatrix} \\ -\frac{2}{10} \begin{pmatrix} 125 \\ 125 \end{pmatrix} + \frac{4}{10} \begin{pmatrix} 125 \\ 125 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 37.5 - 12.5 \\ -25 + 50 \end{bmatrix} = \begin{bmatrix} 25 \\ 25 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 25 \\ 25 \end{bmatrix}$$



$\therefore$ At time $n=2$,

25 machines were infected with malware X
 25 machines were infected with malware Y

c) Find Eigen values + Eigen vectors.

We can find eigenvalues + eigenvectors using the formula

$$\boxed{\det (M - \lambda I) = 0}$$

where $M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (\because M \text{ is a } 2 \times 2 \text{ matrix})$$

$$\underline{M - \lambda I}$$

$$\lambda I = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$M - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\boxed{M - \lambda I = \begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix}}$$

$$\det (M - \lambda I) = ad - bc$$

where ~~a=4~~ $a = 4 - \lambda$ $b = 1$ $c = 2$ $d = 3 - \lambda$

$$\det (M - \lambda I) = (4 - \lambda)(3 - \lambda) - 1(2)$$

$$= 12 - 4\lambda - 3\lambda + \lambda^2 - 2$$

$$\lambda^2 - 7\lambda + 10$$

$$\boxed{\therefore \det (M - \lambda I) = \lambda^2 - 7\lambda + 10}$$

To find eigen values + eigen vectors,

$$\det(M - \lambda I) = 0$$

$$\therefore \lambda^2 - 7\lambda + 10 = 0$$

$$\lambda^2 - 5\lambda - 2\lambda + 10 = 0 \quad -5\overset{10}{\cancel{-2}}$$

$$\lambda(\lambda - 5) - 2(\lambda - 5) = 0$$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\boxed{\lambda = 2 \mid \lambda = 5}$$

$$\boxed{\therefore \text{Eigen values} = 2 + 5}$$



To find eigen vectors,

$$(M - \lambda I)\vec{v} = \vec{0}$$

$$\cancel{M - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}}$$

$$M - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

When $\lambda = 2$,

$$M - 2I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4-2 & 1-0 \\ 2-0 & 3-2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2x + y = 0$$

$$y = -2x$$

$$\boxed{\therefore \bar{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}}$$

when $\lambda = 5$,

$$M - 5I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 4-5 & 1-0 \\ 2-0 & 3-5 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x + y = 0$$

$$y = x$$

$$\boxed{\bar{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

$\therefore$ Eigen vector when $\lambda = 2 \Rightarrow \bar{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

Eigen vector when $\lambda = 5 \Rightarrow \bar{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$



d) Given $m = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

P = Matrix of Eigen vectors.

$$P = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \quad \left(\because \bar{v}_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \bar{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$$

D = Diagonal of Eigen values

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \quad \left(\because \lambda_1 = 2 \quad \lambda_2 = 5 \right)$$

$$P^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

where $a = -1 \quad b = 1 \quad c = 2 \quad d = 1$

$$P^{-1} = \frac{1}{-1-2} \begin{bmatrix} 1 & -1 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

we need to verify $M = PDP^{-1}$

$$PD = \begin{pmatrix} -1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} -2+0 & 0+5 \\ 4+0 & 0+5 \end{pmatrix}$$

$$PD = \begin{pmatrix} -2 & 5 \\ 4 & 5 \end{pmatrix}$$

$$PDP^{-1} = \begin{pmatrix} -2 & 5 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} \frac{-2}{3} + \frac{10}{3} & \frac{-2}{3} + \frac{5}{3} \\ -\frac{4}{3} + \frac{10}{3} & \frac{4}{3} + \frac{5}{3} \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = M$$

$$\therefore M = PDP^{-1}$$

$$\therefore M^n = PD^n P^{-1}$$

$$\therefore M^n = PD^n P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & 5^n \end{bmatrix} \begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$



e) from d, we know that

$$M = P D P^{-1}$$

$$P = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} -1/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix}$$

$$\text{Given } \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\therefore \begin{pmatrix} x_n \\ y_n \end{pmatrix} = M^{n-1} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = P D^{n-1} P^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\boxed{\begin{pmatrix} x_n \\ y_n \end{pmatrix} = P D^{n-1} P^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}}$$

$$P^{-1} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{bmatrix} -1/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} -1/3 \\ 2/3 \end{pmatrix} = \begin{bmatrix} -1/3 \\ 2/3 \end{bmatrix}$$

$$D^{n-1} \begin{pmatrix} -1/3 \\ 2/3 \end{pmatrix} = \begin{bmatrix} 2^{n-1} & 0 \\ 0 & 5^{n-1} \end{bmatrix} \begin{bmatrix} -1/3 \\ 2/3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 (2^{n-1}) \\ 5^{n-1} (2/3) \end{bmatrix} = \begin{bmatrix} \frac{1}{3} (2^{n-1}) \\ \frac{2}{3} (5^{n-1}) \end{bmatrix}$$

$$D^{n-1} P^{-1} = \begin{bmatrix} \frac{1}{3} (2^{n-1}) \\ \frac{2}{3} (5^{n-1}) \end{bmatrix}$$

$$P D^{n-1} P^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} (2^{n-1}) \\ \frac{2}{3} (5^{n-1}) \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{3} (2^{n-1}) + \frac{2}{3} (5^{n-1}) \\ \frac{2}{3} (2^{n-1}) + \frac{2}{3} (5^{n-1}) \end{bmatrix}$$

When n grows larger, 5^{n-1} dominates 2^{n-1} becomes negligible.

~~$$x_n \approx \frac{2}{3} 5^{n-1}$$~~

$$x_n \approx \frac{2}{3} (5^{n-1})$$

$$y_n \approx \frac{2}{3} (5^{n-1})$$

Total number of infected machine

$$x_n + y_n = \frac{2}{3} (5^{n-1}) + \frac{2}{3} (5^{n-1}) = \frac{4}{3} (5^{n-1})$$

Total number of infected machine

$$\text{with } x = \frac{x_n}{x_n + y_n} = \frac{\frac{2}{3} (5^{n-1})}{\frac{4}{3} (5^{n-1})} = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$$



$\therefore$ The numbers of machines infected with x are of the proportion $\frac{1}{2}$

3)

$$(x-6)^2 + (y-18)^2 \leq 361$$

The first circle has a center $(6, 18)$ with radius $\sqrt{361} = 19$.

$$(x+6)^2 + (y+18)^2 \leq 361$$

The second circle has a center $(-6, -18)$ with radius $\sqrt{361} = 19$.

The line equation $x-y=0 \Rightarrow y=x$.

Find intersections of the two circles

$$\begin{aligned} (x-6)^2 + (y-18)^2 &= (x+6)^2 + (y+18)^2 \\ (x-6)^2 + (y-18)^2 &\Rightarrow \\ x^2 - 12x + 36 + y^2 - 36y + 324 &= x^2 + y^2 - 12x - 36y + 360 \end{aligned}$$

$$\begin{aligned} (x+6)^2 + (y+18)^2 &\Rightarrow \\ x^2 + 12x + 36 + y^2 + 36y + 324 &= x^2 + y^2 + 12x + 36y + 360 \end{aligned}$$

$$\therefore x^2 + y^2 - 12x - 36y + 360 = x^2 + y^2 + 12x + 36y + 360$$

$$\begin{aligned} x^2 + y^2 - 12x - 36y + 360 - x^2 - y^2 - 12x - 36y - 360 &= 0 \\ -24x - 72y &= 0 \end{aligned}$$

$$24x + 72y = 0$$

$$\div 24 \Rightarrow \boxed{x + 3y = 0}$$

$$\boxed{x = -3y}$$

Now substitute $x = -3y$ into $(x-6)^2 + (y-18)^2 = 361$.

$$(-3y-6)^2 + (y-18)^2 = 361$$

~~$$(9y^2 + 36y + 36) + (y^2 - 36y + 324) = 361$$~~

$$(-3y-6)^2 + (y-18)^2 = 361$$

$$(9y^2 + 36y + 36) + (y^2 - 36y + 324) = 361$$

$$10y^2 + 360 = 361$$

$$10y^2 = 1$$

$$y^2 = \frac{1}{10}$$

$$y = \pm \frac{1}{\sqrt{10}}$$

$$y = \frac{1}{\sqrt{10}} \text{ or } y = -\frac{1}{\sqrt{10}}$$

Sub y in $x = -3y$.

$$x = -\frac{3}{\sqrt{10}} \text{ or } \frac{3}{\sqrt{10}}$$

∴ Point of intersection of the two circle are

$$\left(-\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right) \left(\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}} \right)$$



Point of each circle intersecting with line $y=x$.

i) $(x-6)^2 + (y-18)^2 = 361$ with line $y=x$.

Sub $y=x$ in $(x-6)^2 + (y-18)^2 = 361$

$$(x-6)^2 + (x-18)^2 = 361$$

$$(x^2 - 12x + 36) + (x^2 - 36x + 324) = 361$$

$$2x^2 - 48x + 360 = 361$$

$$2x^2 - 48x - 1 = 0$$

$$\div 2$$

$$\Rightarrow x^2 - 24x - \frac{1}{2} = 0$$

factorize using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where $a=1$ $b=-24$ $c=-\frac{1}{2}$.

$$x = \frac{-(-24) \pm \sqrt{(-24)^2 - 4(1)(-\frac{1}{2})}}{2(1)}$$

$$= \frac{24 \pm \sqrt{576 + 2}}{2} = \frac{24 \pm \sqrt{578}}{2}$$

$\therefore$ Points of intersection between $(x-6)^2 + (y-18)^2 = 361$ & line $y=x$ is

$$\left(\frac{24 + \sqrt{578}}{2}, \frac{24 + \sqrt{578}}{2} \right) \text{ and } \left(\frac{24 - \sqrt{578}}{2}, \frac{24 - \sqrt{578}}{2} \right)$$


$$17) (x+6)^2 + (y+18)^2 = 361 \text{ with } y=x.$$

$$(x+6)^2 + (y+18)^2 = 361.$$

$$\text{Sub } y=x \text{ in } (x+6)^2 + (y+18)^2$$

$$(x+6)^2 + (x+18)^2 = 361$$

$$x^2 + 12x + 36 + x^2 + 36x + 324 = 361$$

$$2x^2 + 48x + 360 = 361$$

$$2x^2 + 48x - 1 = 0$$

$$x^2 + 24x - \frac{1}{2} = 0$$

$$x = \frac{-24 \pm \sqrt{24^2 - 4(1)(-\frac{1}{2})}}{2(1)}$$

$$= \frac{-24 \pm \sqrt{576+2}}{2}$$

$$= \frac{-24 \pm \sqrt{578}}{2}.$$

$\therefore$ Point of intersection is

$$\left(\frac{-24 + \sqrt{578}}{2}, \frac{-24 + \sqrt{578}}{2} \right) \text{ and } \left(\frac{-24 - \sqrt{578}}{2}, \frac{-24 - \sqrt{578}}{2} \right)$$

Point of intersection with axes

i) Find circle $(x-6)^2 + (y-18)^2 = 361$ with x axis $\Rightarrow y=0$.

Sub $y=0$ in $(x-6)^2 + (y-18)^2 = 361$

$$(x-6)^2 + (0-18)^2 = 361$$

$$(x-6)^2 + 324 = 361$$

$$(x-6)^2 = 37$$

$$x-6 = \pm\sqrt{37}$$

$$x = 6 \pm \sqrt{37}$$

$\therefore$ Point of circle $(x-6)^2 + (y-18)^2 = 361$ with x axis is $(6 + \sqrt{37}, 0)$ $(6 - \sqrt{37}, 0)$

ii) Second circle $(x+6)^2 + (y+18)^2 = 361$ with x axis $\Rightarrow y=0$.

Sub $y=0$ in $(x+6)^2 + (y+18)^2 = 361$

$$(x+6)^2 + (0+18)^2 = 361$$

$$(x+6)^2 + 324 = 361$$

$$(x+6)^2 = 37$$

$$x+6 = \pm\sqrt{37}$$

$$\boxed{x = -6 \pm \sqrt{37}}$$

$\therefore$ Point of circle $(x+6)^2 + (y+18)^2 = 361$ with x axis is $(-6 + \sqrt{37}, 0)$ and $(-6 - \sqrt{37}, 0)$

iii) Second circle $(x-6)^2 + (y-18)^2 = 361$ with y axis
 $\Rightarrow x=0$

$$(0-6)^2 + (y-18)^2 = 361$$

$$(y-18)^2 = 361 - 36$$

$$(y-18)^2 = 325$$

$$y-18 = \pm \sqrt{325}$$

$$\boxed{y = 18 \pm \sqrt{325}}$$

$\therefore$ Point of intersection of $(x-6)^2 + (y-18)^2 = 361$
with y axis is $(0, 18 + \sqrt{325})$ & $(0, 18 - \sqrt{325})$

iv) Second circle $(x+6)^2 + (y+18)^2 = 361$ with y axis
 $x=0$

$$(0+6)^2 + (y+18)^2 = 361$$

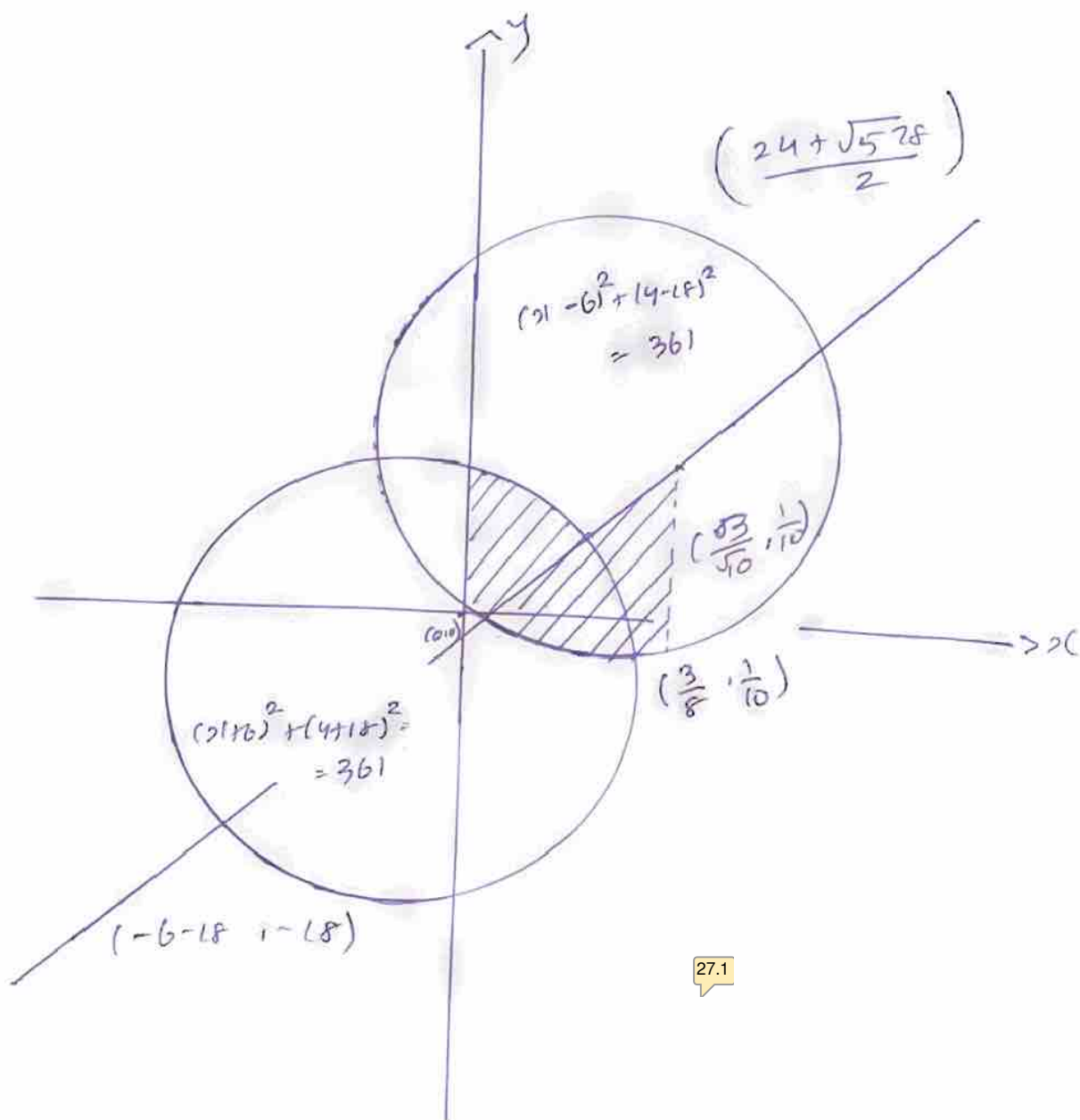
$$(y+18)^2 = 361 - 36$$

$$(y+18)^2 = 325$$

$$y+18 = \pm \sqrt{325}$$

$$\boxed{y = -18 \pm \sqrt{325}}$$

$\therefore$ Point of intersection of $(x+6)^2 + (y+18)^2 = 361$
with y axis is $(0, -18 + \sqrt{325})$ & $(0, -18 - \sqrt{325})$



Index of comments

- 4.1 this is one possible solution, but not all of them
- 27.1 the intersection points are correct, but not the shaded region